QM Honors HW#8 Biao Huang

$\boxed{\text{Total} = 60}$

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15.1.2 (1) $H_{hf} = A\vec{S_1}\cdot\vec{S_2} = \frac{A}{2}(\vec{S}^2 - \vec{S_1}^2 - \vec{S_2}^2)$

$\Rightarrow S = 1$, $S_1 = S_2 = \frac{1}{2}$: $\begin{cases} |++\rangle \\ \frac{1}{\sqrt{2}}(|+-\rangle + |-+\rangle) \\ |--\rangle \end{cases}$ $H_{hf} = \frac{\hbar^2}{2}A(2 - \frac{3}{4} - \frac{3}{4}) = \frac{A\hbar^2}{4}$

$S = 0$, $S_1 = S_2 = \frac{1}{2}$: $\frac{1}{\sqrt{2}}(|+-\rangle - |-+\rangle)$ $H_{hf} = \frac{\hbar^2}{2}A(0 - \frac{3}{4} - \frac{3}{4}) = -\frac{3\hbar^2}{4}A$

$$\Rightarrow H = H_0 + H_{hf} = \begin{cases} E_+ : & -Ry + \frac{A\hbar^2}{4} \\ E_- : & -Ry - \frac{3\hbar^2}{4}A \end{cases}$$

(2) $\mathcal{H}_{hf} \approx \frac{\mu_e \cdot \mu_p}{a_0^3} = \gamma_e\gamma_p \frac{\vec{S_1}\cdot\vec{S_2}}{a_0^3} = \frac{g_1 g_2 e^2}{4Mmc^2 a_0^3} \vec{S_1}\cdot\vec{S_2}$

$\boxed{\gamma = \frac{g}{2mc}}$ $A = \frac{2e}{2mc}\frac{5.6e}{2Mc}\frac{1}{a_0^3}$ $\boxed{a_0 = \frac{\hbar^2}{me^2}}$

$$\Delta E = E_+ - E_- = (-Ry + \frac{A\hbar^2}{4}) - (-Ry - \frac{3\hbar^2 A}{4}) = A\hbar^2$$

$\boxed{\alpha = \frac{e^2}{\hbar c} = \frac{1}{137}}$ $= \frac{5.6}{M}\frac{e^2\hbar^2}{2mc^2}\frac{m^3e^6}{\hbar^6} = \frac{5.6}{M} Ry \frac{e^4 m}{\hbar^2 c^2}$

$\boxed{Ry = \frac{me^4}{2\hbar^2} = 13.6eV}$ $= 5.6\frac{m}{M} Ry\, \alpha^2$

$$\Rightarrow \omega = \frac{\Delta E}{\hbar} = 5.6 \frac{mc^2}{Mc^2} \frac{Ry\, \alpha^2}{\hbar}$$

$$\cong 5.6 \frac{0.511}{938} \frac{13.6 \times (1/137)^2 \times 1.6 \times 10^{-19}}{\left(\frac{6.64}{2\times 3.14}\right) \times 10^{-34}} s^{-1}$$

$$\cong 3.35 \times 10^{9} \ s^{-1}$$

$mc^2 = 0.511$ MeV
$Mc^2 = 938.28$ MeV
$Ry = 13.4$ eV
$\hbar c = 1973.3$ eV Å
$\alpha = 1/137$

$$\Rightarrow \lambda = \frac{2\pi c}{\omega} = 0.56 \ m$$ (If use $\frac{m}{M} \sim \frac{1}{2000}$, $\lambda \sim 0.61 m$)

three states, each with prob. $e^{-\beta E+}$

$\approx 7.8 \times 10^{-5}$

(3) $$\frac{P(\text{triplet})}{P(\text{singlet})} = \frac{3e^{-\beta E+}}{e^{-\beta E-}} = 3e^{-\beta A \hbar^2} \approx 3$$

according to Boltzmann distribution

(20)
15.2.2 CG coefficients

① $\frac{1}{2} \otimes 1 = \frac{3}{2} \oplus \frac{1}{2}$

$m \downarrow$ $j \rightarrow$	$\frac{3}{2}$	$\frac{1}{2}$
$\frac{3}{2}$	$\|\frac{3}{2}, \frac{3}{2}\rangle$	
$\frac{1}{2}$	$\|\frac{3}{2}, \frac{1}{2}\rangle$	$\|\frac{1}{2}, \frac{1}{2}\rangle$
$-\frac{1}{2}$	$\|\frac{3}{2}, -\frac{1}{2}\rangle$	$\|\frac{1}{2}, -\frac{1}{2}\rangle$
$-\frac{3}{2}$	$\|\frac{3}{2}, -\frac{3}{2}\rangle$	

First colomn :

- $\boxed{\left|\frac{3}{2}, \frac{3}{2}\right\rangle} = \left|1\,1, \frac{1}{2}\frac{1}{2}\right\rangle$

- $J_-\left|\frac{3}{2}, \frac{3}{2}\right\rangle = \hbar\sqrt{3}\left|\frac{3}{2}, \frac{1}{2}\right\rangle$ $\qquad (J_\pm|jm\rangle = \sqrt{(j\mp m)(j\pm m+1)}\,\hbar\,|j, m\pm 1\rangle)$

$$\Rightarrow \boxed{\left|\frac{3}{2}, \frac{1}{2}\right\rangle} = \frac{J_-}{\sqrt{3}\hbar}\left|\frac{3}{2}, \frac{3}{2}\right\rangle$$

$$= \frac{1}{\sqrt{3}\hbar}(J_{1-} + J_{2-})\left|1\,1, \frac{1}{2}\frac{1}{2}\right\rangle$$

$$= \frac{1}{\sqrt{3}\hbar}\left[\hbar\left|1\,1, \frac{1}{2}(-\tfrac{1}{2})\right\rangle + \hbar\sqrt{2}\left|1\,0, \frac{1}{2}\frac{1}{2}\right\rangle\right]$$

$$\boxed{= \frac{1}{\sqrt{3}}\left|1\,1, \frac{1}{2}(-\tfrac{1}{2})\right\rangle + \sqrt{\frac{2}{3}}\left|1\,0, \frac{1}{2}\frac{1}{2}\right\rangle}$$

Now use $\langle j_1(-m_1), j_2(-m_2)|j(-m)\rangle = (-1)^{j_1+j_2-j}\langle j_1 m_1, j_2 m_2|jm\rangle$,

- $\boxed{\left|\frac{3}{2}, -\frac{1}{2}\right\rangle} = \frac{(-1)^{\frac{1}{2}+1-\frac{3}{2}}}{\sqrt{3}}\left|1(-1), \frac{1}{2}\frac{1}{2}\right\rangle + \sqrt{\frac{2}{3}}(-1)^{\frac{1}{2}+1-\frac{3}{2}}\left|1\,0, \frac{1}{2}(-\tfrac{1}{2})\right\rangle$

$$\boxed{= \frac{1}{\sqrt{3}}\left|1(-1), \frac{1}{2}\frac{1}{2}\right\rangle + \sqrt{\frac{2}{3}}\left|1\,0, \frac{1}{2}(-\tfrac{1}{2})\right\rangle}$$

- $\boxed{\left|\frac{3}{2}, -\frac{3}{2}\right\rangle} = (-1)^{\frac{1}{2}+1-\frac{3}{2}}\left|1(-1), \frac{1}{2}(-\tfrac{1}{2})\right\rangle = \boxed{\left|1(-1), \frac{1}{2}(-\tfrac{1}{2})\right\rangle}$

Second colomn

- $|\frac{1}{2}\ \frac{1}{2}\rangle \sim \begin{cases} a) \sim |\frac{3}{2}, \frac{1}{2}\rangle \sim |10, \frac{1}{2}\frac{1}{2}\rangle \ \& \ |11, \frac{1}{2}(-\frac{1}{2})\rangle \\ b) \perp \ |\frac{3}{2}, \frac{1}{2}\rangle = \frac{1}{\sqrt{3}}|11, \frac{1}{2}(-\frac{1}{2})\rangle + \sqrt{\frac{2}{3}}\,|10, \frac{1}{2}\frac{1}{2}\rangle \\ c) \text{ Normalized} \end{cases}$

$$\Rightarrow \boxed{|\tfrac{1}{2}, \tfrac{1}{2}\rangle = \frac{\sqrt{2}}{\sqrt{3}}|11, \tfrac{1}{2}(-\tfrac{1}{2})\rangle - \frac{1}{\sqrt{3}}|10, \tfrac{1}{2}\tfrac{1}{2}\rangle}$$

- $\boxed{|\frac{1}{2}, -\frac{1}{2}\rangle} = (-1)^{\frac{1}{2}+1-\frac{1}{2}}\left[\frac{\sqrt{2}}{\sqrt{3}}|1(-1), \frac{1}{2}\frac{1}{2}\rangle - \frac{1}{\sqrt{3}}|10, \frac{1}{2}(-\frac{1}{2})\rangle\right]$

$$\boxed{= \frac{1}{\sqrt{3}}|10, \tfrac{1}{2}(-\tfrac{1}{2})\rangle - \frac{\sqrt{2}}{\sqrt{3}}|1(-1), \tfrac{1}{2}\tfrac{1}{2}\rangle}$$

② $1 \otimes 1 = 2 \oplus 1 \oplus 0$

m \ j →	2	1	0
2	$\|2,2\rangle$		
1	$\|2,1\rangle$	$\|1,1\rangle$	
0	$\|2,0\rangle$	$\|1,0\rangle$	$\|0,0\rangle$
−1	$\|2,-1\rangle$	$\|1,-1\rangle$	
−2	$\|2,-2\rangle$		

$\boxed{j=2}$

- $\boxed{|2,2\rangle = |11,11\rangle}$

- $J_-|2,2\rangle = \hbar\sqrt{4}\,|2,1\rangle$

$$\Rightarrow \boxed{|2,1\rangle} = \frac{1}{2\hbar}(J_{1-}+J_{2-})|11,11\rangle$$

$$= \frac{\hbar}{2\hbar}\left[\sqrt{2}|10,11\rangle + \sqrt{2}|11,10\rangle\right]$$

$$\boxed{= \frac{1}{\sqrt{2}}\left(|10,11\rangle + |11,10\rangle\right)}$$

- $J_-|21\rangle = \hbar\sqrt{3\times 2}\,|2,0\rangle$

$$\boxed{|2,0\rangle} = \frac{1}{\sqrt{6}\hbar}(J_{1-}+J_{2-})|2,1\rangle$$

$$= \frac{\hbar}{\sqrt{6}\hbar\sqrt{2}}\left[\sqrt{2}|1(-1),11\rangle + \sqrt{2}|10,10\rangle + \sqrt{2}|10,10\rangle + \sqrt{2}|11,1(-1)\rangle\right]$$

$$\boxed{= \frac{1}{\sqrt{6}}\left[|11,1(-1)\rangle + 2|10,10\rangle + |1(-1),11\rangle\right]}$$

- $\boxed{|2,-1\rangle} = (-1)^{1+1-2}\frac{1}{\sqrt{2}}\left[|10,1(-1)\rangle + |1(-1),10\rangle\right]$

$$\boxed{= \frac{1}{\sqrt{2}}\left[|10,1(-1)\rangle + |1(-1),10\rangle\right]}$$

- $\boxed{|2,-2\rangle} = (-1)^{1+1-2}\,|1(-1),1(-1)\rangle = \boxed{|1(-1),1(-1)\rangle}$

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$\boxed{j=1}$

- $|11\rangle \sim \begin{cases} |1 1, 1 0\rangle \ \& \ |1 0, 1 1\rangle \\ \perp \ |2, 1\rangle = \frac{1}{\sqrt{2}}\left(|1 0, 1 1\rangle + |1 1, 1 0\rangle\right) \\ \text{Normalized} \end{cases}$

$$\Rightarrow \boxed{|11\rangle = \frac{1}{\sqrt{2}}\left(|1 1, 1 0\rangle - |1 0, 1 1\rangle\right)}$$

- $J_- |11\rangle = \hbar\sqrt{2}\,|10\rangle$

$$\Rightarrow \boxed{|10\rangle} = \frac{1}{\sqrt{2}\hbar}\left(J_{1-} + J_{2-}\right)|11\rangle$$

$$= \frac{1}{\sqrt{2}\hbar}\frac{\hbar}{\sqrt{2}}\left(\sqrt{2}\,|10, 10\rangle - \sqrt{2}\,|1 -1, 11\rangle + \sqrt{2}\,|11, 1 -1\rangle - \sqrt{2}\,|10, 10\rangle\right)$$

$$\boxed{= \frac{1}{\sqrt{2}}\left(|11, 1 -1\rangle - |1 -1, 11\rangle\right)}$$

- $\boxed{|1 -1\rangle} = (-1)^{1+1-1}\frac{1}{\sqrt{2}}\left(|1 -1, 10\rangle - |10, 1 -1\rangle\right)$

$$\boxed{= \frac{1}{\sqrt{2}}\left(|10, 1 -1\rangle - |1 -1, 10\rangle\right)}$$

$\boxed{j=0}$ $\quad |00\rangle \sim \begin{cases} \sim |11, 1 -1\rangle, |10, 10\rangle, |1 -1, 11\rangle \\ \perp |10\rangle = \frac{1}{\sqrt{2}}\left(|11, 1 -1\rangle - |1 -1, 11\rangle\right) \\ \perp |20\rangle = \frac{1}{\sqrt{6}}\left(|11, 1 -1\rangle + 2|10, 10\rangle + |1 -1, 11\rangle\right) \\ \text{Normalized} \end{cases}$

$$\Rightarrow \boxed{|00\rangle = \frac{1}{\sqrt{3}}\left[|11, 1 -1\rangle - |10, 10\rangle + |1 -1, 11\rangle\right]}$$

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15.2.5 ① $P_1 P_0 = \frac{3}{16} I - \frac{1}{2\hbar^2}\vec{S}_1\cdot\vec{S}_2 - \frac{1}{\hbar^4}(\vec{S}_1\cdot\vec{S}_2)^2$

Here $(\vec{S}_1\cdot\vec{S}_2)^2 = \frac{\hbar}{4}^2 S_1^2 I + i\underbrace{(\vec{S}_1\times\vec{S}_1)}_{= i\hbar\vec{S}_1}\cdot\vec{S}_2\,\frac{\hbar}{2} = \frac{3\hbar^4}{16} I - \frac{\hbar^2}{2}\vec{S}_1\cdot\vec{S}_2$ $\Rightarrow$

where $(\vec{A}\cdot\vec{\sigma})(\vec{B}\cdot\vec{\sigma}) = \vec{A}\cdot\vec{B}\, I + i(\vec{A}\times\vec{B})\cdot\vec{\sigma}$ has been exploited

$$\Rightarrow P_1 P_0 = 0$$

$$P_1^2 = \frac{9}{16} I + \frac{3}{2\hbar^2}\vec{S}_1\cdot\vec{S}_2 + \frac{1}{\hbar^4}(\vec{S}_1\cdot\vec{S}_2)^2$$

$$= \frac{3}{4} I + \vec{S}_1\cdot\vec{S}_2\,\frac{1}{\hbar^2} = P_1$$

$$P_2^2 = \frac{1}{16} I - \frac{1}{2\hbar^2}\vec{S}_1\cdot\vec{S}_2 + \frac{1}{\hbar^4}(\vec{S}_1\cdot\vec{S}_2)^2 = \frac{1}{4} I - \frac{1}{\hbar^2}\vec{S}_1\cdot\vec{S}_2 = P_2$$

$$\Rightarrow P_i P_j = \delta_{ij} P_j$$

② $\vec{S}_1\cdot\vec{S}_2 = \frac{1}{2}(\vec{S}^2 - \vec{S}_1^2 - \vec{S}_2^2) = \frac{1}{2}\vec{S}^2 - \frac{1}{2}\,\frac{1}{2}\times\frac{3\hbar^2}{2} - \frac{1}{2}\,\frac{1}{2}\times\frac{3}{2}\hbar^2$

$$= \frac{1}{2}\vec{S}^2 - \frac{3}{4}\hbar^2 I$$

as in $\frac{1}{2}\otimes\frac{1}{2}$, $S_1 = S_2 = \frac{1}{2}$, $\vec{S}_i^2 = S_i(S_i+1)\hbar = \frac{1}{2}\times\frac{3}{2}\hbar$

$$\Rightarrow P_1 = \frac{3}{4} I + \frac{1}{2\hbar^2}\vec{S}^2 - \frac{3}{4} I = \frac{1}{2\hbar^2}\vec{S}^2$$

$\Rightarrow P_1 |1,m\rangle = \frac{1}{2\hbar^2}\, 1\times 2\hbar^2 |1,m\rangle = |1,m\rangle$

$P_1 |0,m\rangle = \frac{1}{2\hbar^2}\, 0\times 1\,\hbar^2 |0,m\rangle = 0$

P_1 projects into $S=1$ space

$$P_0 = \frac{1}{4}I - \frac{1}{\hbar^2}\left(\frac{1}{2}\vec{S}^2 - \frac{3}{4}\hbar^2 I\right) = I - \frac{1}{2\hbar^2}\vec{S}^2$$

$$\Rightarrow \quad P_0|1m\rangle = |1m\rangle - \frac{1}{2\hbar^2}\,1\times 2\hbar^2|1m\rangle = 0$$

$$P_0|0m\rangle = |0m\rangle - \frac{1}{2\hbar^2}\,0\times 1\hbar^2|0m\rangle = |0m\rangle$$

$\Rightarrow$ P_0 projects into $S=0$ space

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15.2.7 $\quad j_1 \otimes j_1 = 2j_1 \oplus 2j_1 - 1 \oplus \cdots \oplus 0$

- $|2j_1, 2j_1\rangle = |j_1 j_1, j_1 j_1\rangle$

$$J_-|2j_1, 2j_1\rangle = 2\sqrt{j_1}\,\hbar\,|2j_1, 2j_1-1\rangle$$

$$|2j_1, 2j_1-1\rangle = \frac{1}{2\hbar\sqrt{j_1}}(J_{1-} + J_{2-})|j_1 j_1, j_1 j_1\rangle$$

$$= \frac{1}{2\hbar\sqrt{j_1}}\left[\sqrt{2j_1}\,\hbar|j_1 j_1-1, j_1 j_1\rangle + \sqrt{2j_1}\,\hbar|j_1 j_1, j_1 j_1-1\rangle\right)$$

$$= \frac{1}{\sqrt{2}}\left[|j_1 j_1-1, j_1 j_1\rangle + |j_1 j_1, j_1 j_1-1\rangle\right].$$

We notice that in $J_- = J_{1-} + J_{2-}$, J_{1-}, J_{2-} will not produce negative sign when acting on $J_{1-}|j_1 m_1, j_2 m_2\rangle = \sqrt{(j_1+m_1)(j_1-m_1+1)}\,\hbar\,|j_1 m_1-1, j_2 m_2\rangle$ (and similar for J_{2-}). Thus, all terms will be positive.

Further, $J_- = J_{1-} + J_{2-}$, J_{1-} & J_{2-} will act on states together everytime, producing the same coefficients and lowering of m_1, m_2.

Thus, in states $|2j_1, m\rangle \sim \{ |j_1 m_1, j_1 m_2\rangle \}$, m_1, m_2 are symmetric. J_- does not change the symmetry between m_1, m_2

On the other hand,

$$|2j_1-1, 2j_1-1\rangle \sim \begin{cases} \sim |j_1 j_1-1, j_1 j_1\rangle,\ |j_1 j_1, j_1 j_1-1\rangle \\ \perp\ |2j_1, 2j_1-1\rangle = \frac{1}{\sqrt{2}}\left[|j_1 j_1-1, j_1 j_1\rangle + |j_1 j_1, j_1 j_1-1\rangle \right] \\ \text{Normalized.} \end{cases}$$

$$\Rightarrow\ |2j_1-1, 2j_1-1\rangle = \frac{1}{\sqrt{2}}\left[|j_1 j_1, j_1 j_1-1\rangle - |j_1 j_1-1, j_1 j_1\rangle \right]$$

which is anti-symmetric. Since $J_- = J_{1-} + J_{2-}$ has the same action on m_1, m_2, again, we have that in

$$|2j_1-1, m\rangle \sim \{ |j_1 m_1, j_1 m_2\rangle \},$$

terms involving m_1, m_2 are antisymmetric

10 5 Particle in a box

16.1.2 $\psi = (x-a)(x+a) = x^2 - a^2$, $H = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2}$

$$E[\psi] = \frac{\int_{-a}^{a} (x^2-a^2)\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\right)(x^2-a^2)\,dx}{\int_{-a}^{a} (x^2-a^2)^2\,dx}$$

$$= -\frac{\hbar^2}{m}\frac{\int_{-a}^{a} (x^2-a^2)\,dx}{\int_{-a}^{a} (x^4 - 2a^2x^2 + a^4)\,dx}$$

$$= -\frac{\hbar^2}{m}\;\frac{\frac{2}{3}a^3 - 2a^3}{\frac{2}{5}a^5 - \frac{4a^5}{3} + 2a^5} \quad \begin{matrix} = -\frac{4}{3} \\ = \frac{16}{15} \end{matrix} \quad = \frac{5}{4a^2}$$

$$= \frac{\hbar^2}{8ma^2}\cdot 10 \geq E_0$$

True $E_0 = \frac{\hbar^2\pi^2}{8ma^2}$ (notice here $L \to 2L$ compared with (5.2.17c) in Shankar)

Thus, the estimated energy is $\frac{10}{\pi^2}$ times of E_0

10

16.1.4 Oscillator $H = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{1}{2}m\omega^2 x^2$

$$\psi = \begin{cases} (x-a)(x+a)^2 & |x| \le a \\ 0 & |x| > a \end{cases}$$

$$E[\psi] = \frac{\int_{-a}^{a} (x^2-a^2)^2\left[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{1}{2}m\omega^2x^2\right](x^2-a^2)^2\,dx}{\int_{-a}^{a}(x^2-a^2)^4\,dx}$$

$$= \frac{\int_{-a}^{a} (x^2-a^2)^2\left[-\frac{\hbar^2}{2m}4(3x^2-a^2) + \frac{1}{2}m\omega^2x^2(x^2-a^2)^2\right]dx}{\int_{-a}^{a}(x^2-a^2)^4\,dx}$$

See attached Mathematica file $= \frac{3\hbar^2}{2a^2m} + \frac{1}{22}m\omega^2a^2$

$$\delta E[\psi] = \left[-\frac{3\hbar^2}{m}\frac{1}{a^3} + \frac{1}{11}m\omega^2 a\right]\delta a = 0$$

$$\Rightarrow \quad a = \left(\frac{33\hbar^2}{m^2\omega^2}\right)^{\frac{1}{4}} = (33)^{\frac{1}{4}}\sqrt{\frac{\hbar}{m\omega}}$$

$$\Rightarrow E[\psi]_{min} = \hbar\omega\left[\frac{1}{2}\sqrt{\frac{3}{11}} + \frac{1}{2}\sqrt{\frac{3}{11}}\right] = \sqrt{\frac{3}{11}}\,\hbar\omega \ge E_0.$$

~ 0.522

True $E_0 = \frac{\hbar\omega}{2} = 0.5\hbar\omega$

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In[31]:= Energy =
  Integrate[(x^2 - a^2)^2 * (-hbar^2/2/m * 4 * (3 * x^2 - a^2) +
       1/2 * m * w^2 * x^2 * (x^2 - a^2)^2), {x, -a, a}] /
   Integrate[(x^2 - a^2)^4, {x, -a, a}]
```

Out[31]= $$\frac{315\left(\frac{128\,a^7\,\text{hbar}^2}{105\,m}+\frac{128\,a^{11}\,m\,w^2}{3465}\right)}{256\,a^9}$$

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In[32]:= Simplify[%]
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Out[32]= $$\frac{3\,\text{hbar}^2}{2\,a^2\,m}+\frac{1}{22}\,a^2\,m\,w^2$$

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In[35]:= N[Sqrt[3/11]]
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Out[35]= 0.522233